



**PROJECT NAME: Automatic Solar Tracking Based
Mobile Charging and Lighting System**

**PROJECT COURSE NO: ECE 3200
COURSE NAME: Electronics Project Design/Development**

Submitted To: • Naymur Rahman Assistant Professor, Department of Electronics and Communication Engineering KUET	Submitted by: • Rukaiya Alam Nidra Roll – 2109019 • Debraj Kundu Roll – 2109052
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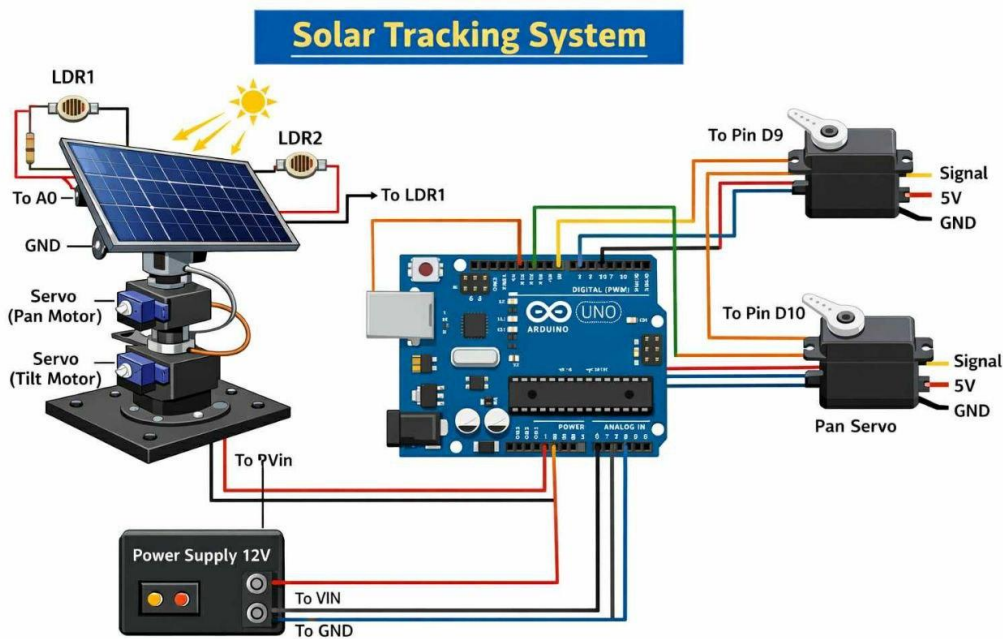
1.Introduction:

In the pursuit of sustainable energy solutions, solar power stands out as a clean, renewable, and widely accessible resource. However, conventional solar panels often suffer from suboptimal energy capture due to static positioning. This project proposes an integrated system that combines automatic solar tracking with mobile charging and lighting functionalities. By leveraging Arduino-based control, dual-axis tracking, and efficient power conversion, the system maximizes solar energy utilization for off-grid applications such as rural electrification, emergency lighting, and portable device charging.

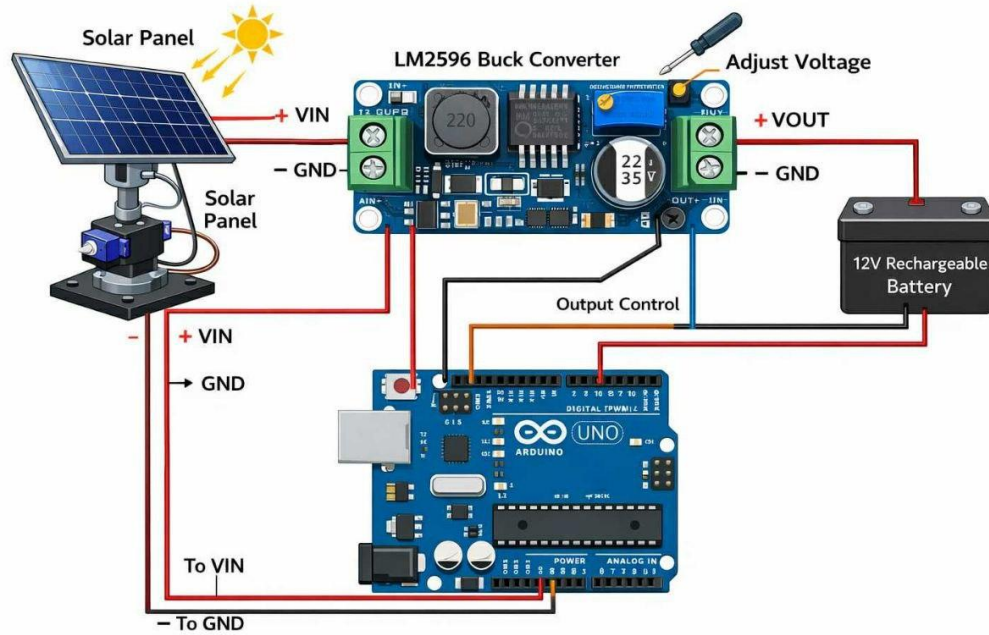
2.Components:

Component	Function
Solar Panel	Primary source of renewable energy
Arduino Uno	Central microcontroller for tracking and control
LDR Sensors (x2)	Detect sunlight intensity for directional tracking
Dual Servo Motors	Enable pan and tilt movement of the solar panel
LM2596 Buck Converter	Regulates voltage from solar panel to safely charge battery
12V Rechargeable Battery	Stores energy for later use
DC-AC Inverter	Converts stored DC power to AC for lighting applications
AC Light Bulb	Demonstrates lighting output capability
Mobile Charging Port	Enables USB-based device charging
Wiring and Connectors	Electrical interconnections among components

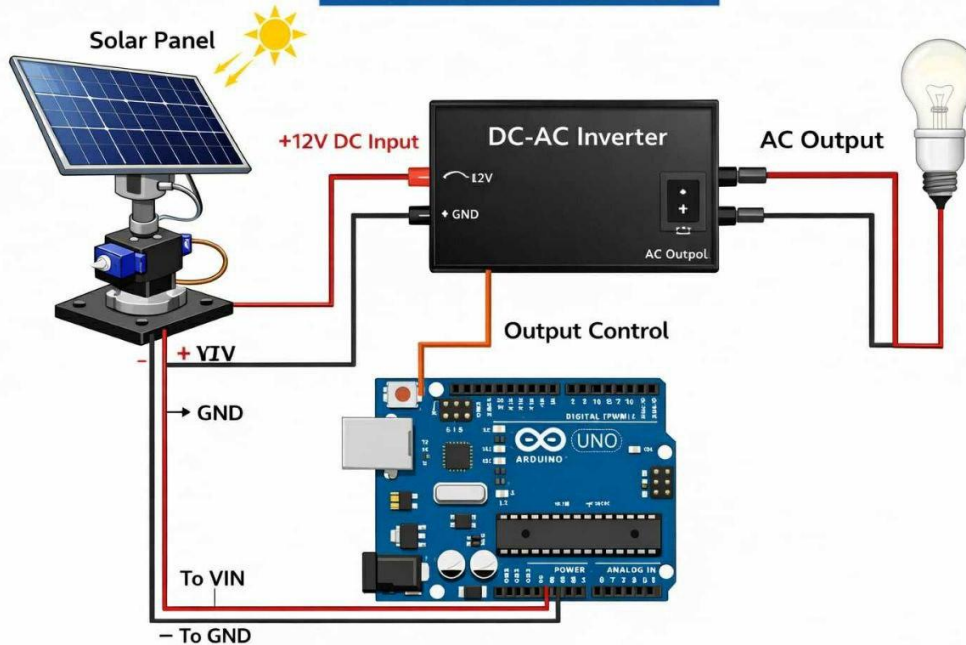
3. Circuit Diagram:



DC-DC Converter



DC-AC Inverter



4.System Overview and Working Principle:

The system operates in three integrated stages:

1. Solar Tracking Mechanism

- Two LDRs detect sunlight intensity.
- Arduino processes LDR data to control dual servo motors.
- The solar panel adjusts its orientation (pan and tilt) to face the sun directly, maximizing energy capture.

2. DC-DC Conversion and Battery Charging

- Solar panel output is regulated via LM2596 Buck Converter.
- The regulated voltage charges a 12V rechargeable battery safely and efficiently.

3. DC-AC Inversion and Load Control

- Stored DC power is converted to AC using a DC-AC inverter.
- Arduino controls inverter output to power an AC light bulb and USB mobile charging port.

5.Possible Outcome:

- **Efficient Solar Tracking:** Enhanced energy capture through dynamic panel orientation.
- **Stable Power Supply:** Reliable voltage regulation and battery storage.
- **Dual Output Capability:** Simultaneous support for AC lighting and DC mobile charging.
- **Smart Control:** Arduino-based automation for tracking and load management.

6. Advantage:

- **Energy Optimization:** Maximizes solar energy harvesting throughout the day.
- **Portability:** Suitable for mobile or remote applications.
- **Cost-Effective:** Uses readily available components and open-source platforms.
- **Eco-Friendly:** Reduces dependence on fossil fuels and grid electricity.
- **Scalability:** Can be expanded for larger systems or integrated with IoT monitoring.

7. Conclusion:

This project demonstrates a practical and intelligent approach to renewable energy utilization. By integrating solar tracking with regulated power conversion and dual output control, it offers a versatile solution for mobile charging and lighting needs. The system is particularly beneficial in rural, disaster-prone, or off-grid environments where reliable energy access is crucial.

8. Future Work:

- **IoT Integration:** Remote monitoring and control via mobile app or web dashboard.
- **MPPT Enhancement:** Incorporate Maximum Power Point Tracking for improved efficiency.
- **Weather Adaptation:** Add sensors for rain and cloud detection to optimize tracking logic.
- **Load Expansion:** Support additional devices like fans, radios, or emergency alarms.
- **Enclosure Design:** Develop a compact, weatherproof casing for field deployment.